

Week 7, Lecture 1: How Sure Are We? Uncertainty from the Shape of the Likelihood

Instructor notes. Part 1: prep reading. Part 2: podium cheat sheet

How to prepare (for students)

- **Big picture.** Last week we found the single best-supported parameter value (the MLE). A point estimate alone doesn't tell us much. Today's question: how uncertain is a parameter estimate, and how do we turn that uncertainty into a confidence interval?
- **Read:** Bolker Ch. 6 through section 6.5 (printed p. 227-267; stop at 6.6, "Comparing models"). Focus on 6.4 (p. 249-260) and 6.5 (p. 260-267). Skim the Bayesian parts (6.2.2, p. 237-243, and 6.4.2, p. 258-260) for the main idea; Wednesday's lecture goes into Bayesian inference in more detail. Bring questions about the reading.
- **Review:** the Poisson distribution and its likelihood (last week's pika haypile example); log rules; derivatives, including second derivatives and the derivative of $\log x$ and of $1/x$.
- **Optional:** look up what a chi-squared (χ^2) distribution is. You don't need its formula, only a rough idea of its shape.

Overview (instructor only)

This file has two parts, both instructor-only. **Part 1** is read beforehand. **Part 2** (after the page break, large print) is the podium copy.

Beat	Time
WOD: <i>Reading the Interval</i> (in class)	~8 min
Lecture	~21 min (as revised)
Discussion	~7 min

As taught (Mon., Oct. 5). The original plan had about twice what fits in the lecture beat. Only sections 1-4 below (big picture, reading questions, curvature, the likelihood ratio interval) were covered. The rest moved:

- Bayesian preview (prior $\times$ likelihood on the pikas): to Wednesday lecture notes (the notes for Week 7, Lecture 2).

- Quadratic approximation, the Hessian, and slices vs. profiles with more than one parameter: to the opening of Lab 6 (see the Lab 6 notes).

Source material: Bolker 6.4.1 (likelihood curves and the likelihood ratio test); legacy `LECTURE4.Rmd` (“Estimating parameter uncertainty”).

Example choice. The one-parameter thread reuses last Monday’s pika haypile data (Week 6, Lecture 1), so students see a familiar likelihood curve and only the new idea (its *width*) is unfamiliar. I avoided the frog chorus data from Lab 5 Act 2 on purpose: if Act 2 went home as a take-home, showing its fitted $\hat{a}, \hat{b}$ on Monday would give away 2b before it’s due.

Where this sits in the week

- **Last week:** likelihood, the NLL, and the MLE (Monday); Lab 5 found an MLE by calculus (Act 1) and with `optim()` (Act 2). Lab 5 Act 1 Part 3 already showed two NLL curves, 3 vs. 30 wetlands, with the same $\hat{p} = 0.4$ but different widths, and asked which survey says more about p . Today formalizes that answer.
- **Today:** the width of the NLL curve becomes a likelihood ratio confidence interval.
- **Tomorrow (Lab 6):** Bliss’s beetle fumigation data. Act 1 (paper) draws a slice and a profile on a very tilted (a, b) surface and uses a 2×2 Hessian to get an SE by hand. Act 2 fits the LD50 directly and compares Hessian and profile intervals. The lab now opens with the quadratic approximation and slices vs. profiles (moved from today).
- **Wednesday:** Bayesian inference, including the pika prior $\times$ likelihood preview moved from today. Wednesday’s take-home WOD, *Curvature by Hand*, has students derive the Poisson and binomial curvatures themselves.

WOD: Reading the Interval (~8 min)

File: `wod/WOD_Week7_Lecture1_ReadTheInterval.Rmd`. Deliberately light (no algebra), since students have had several math-heavy WODs in a row. It gives students the “2 units” rule of thumb *before* the lecture explains it, so section 4 can pay it off.

Chytrid swabs: 2 positives in 20 frogs, binomial, printed NLL-minus-min curve. About 4 min alone, 2 min with a neighbor, 2 min debrief.

- **1a.** $\hat{p} = 0.10$.
- **1b.** Within 2 units: about 0.02 to 0.28 (exact: 0.017 to 0.282).
- **1c.** Not symmetric: 0.08 below $\hat{p}$, 0.18 above. The curve shoots up near 0 because prevalence can’t be negative and 2 positives already rule out values very close to 0; high values are ruled out more gradually.
- **2a.** 8 of 80: same $\hat{p} = 0.10$, curve 4 times steeper, range about 0.05 to 0.18 (narrower, and less lopsided).

Debrief lightly: confirm 1a and 1b, take one answer to 1c, then say “the rest of today explains where the 2 comes from and why the curve’s width matters.” Save the full asymmetry point

for section 4, and don't mention the Wald interval's negative lower bound for these data: that's the payoff of 2d in Wednesday's take-home.

Part 1: Lecture content

1. Big picture (~3 min)

Put the loop on the board:

THEORY $\rightarrow$ MODEL(S) $\rightarrow$ DATA (fit) $\rightarrow$ COMPARE $\rightarrow$ repeat

- So far in the “fit” step we have produced one number per parameter, the MLE. Bolker (p. 249): point estimates are useless without a measure of uncertainty.
- Today's question: *which parameter values are still plausible given the data?* Answer: all the information is in the shape of the likelihood curve near its peak. A narrow valley means the data rule out most values; a broad valley means they don't.
- The same likelihood feeds two schools of inference. Frequentists use it to build confidence intervals (today's focus). Bayesians combine it with a prior to get a probability distribution for the parameter (Wednesday).
- Popperian tie-in, worth saying out loud: a confidence interval is the set of parameter values the data *fail to reject*. Values outside it are falsified (at the 5% level); values inside survive. None is proven, including the MLE.

2. Questions about the reading (~4 min)

Open the floor. If nobody bites, seed with one of these:

- “Bolker says slices are ‘usually wrong.’ Why would fixing the other parameters at their best values be a problem?” (Answer comes in tomorrow's lab.)
- “Why does doubling the data make the *maximum* likelihood smaller but the estimate better?” (Bolker's coin footnote, p. 254: 1 head in 2 flips has probability 0.5; 500 in 1000 has probability 0.025. That doesn't make the coin less fair.)
- “Where does the 1.92 come from?” (Section 4.)

Don't answer at length now if the question is covered later; say “hold that, it's coming in ten minutes” and write it in the corner of the board so you remember to close the loop.

3. Curvature is information (~6 min)

Callback: Lab 5 Act 1, question 3a. Same $\hat{p}$, but the 30-wetland curve was narrower. Now do the same thing with the pikas. Imagine the biologist had surveyed 24 plots, with the same mean (each of the six counts appearing four times).

```
haypiles <- c(2, 5, 3, 0, 4, 4)
n_plots <- length(haypiles)
```

```

lambda_hat <- mean(haypiles)

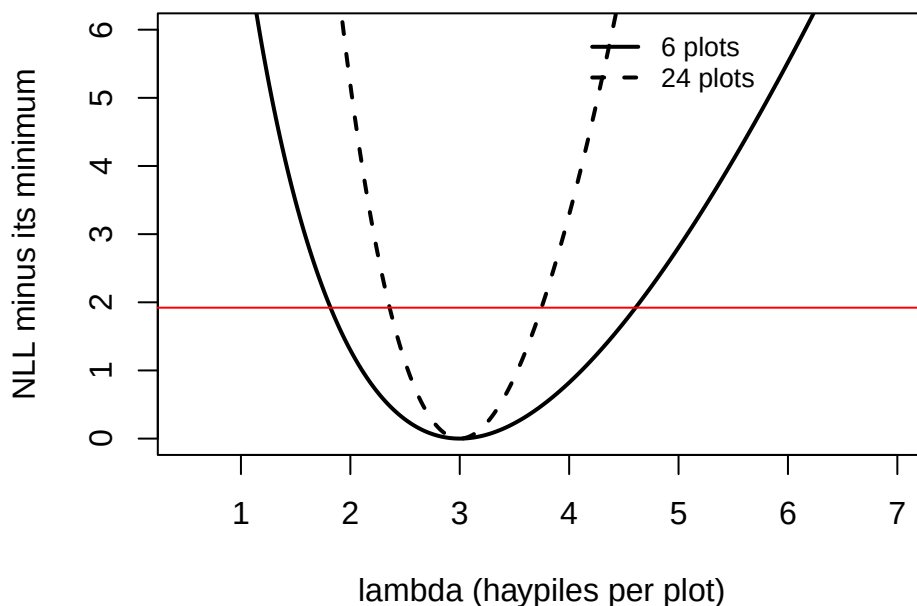
nll_pois <- function(lambda, y) {
  -sum(dpois(y, lambda = lambda, log = TRUE))
}

lambda_grid <- seq(0.01, 8, by = 0.01)

haypiles_24 <- rep(haypiles, 4)
dnll_6 <- sapply(lambda_grid, nll_pois, y = haypiles) - nll_pois(3, haypiles)
dnll_24 <- sapply(lambda_grid, nll_pois, y = haypiles_24) - nll_pois(3, haypiles_24)
lr_cut <- qchisq(0.95, df = 1) / 2

par(mar = c(4, 4.5, 0.5, 0.5))
plot(lambda_grid, dnll_6, type = "l", lwd = 2, xlim = c(0.5, 7), ylim = c(0, 6),
      xlab = "lambda (haypiles per plot)", ylab = "NLL minus its minimum")
lines(lambda_grid, dnll_24, lwd = 2, lty = 2)
abline(h = lr_cut, col = "red")
legend(x = 4.0, y = 6.2, c("6 plots", "24 plots"), lwd = 2, lty = c(1, 2),
      bty = "n", cex = 0.8)

```



- Both bottom out at $\lambda = 3$.
- The NLL is a *sum* over observations, so 4 times the data means 4 times the NLL, everywhere. The valley gets 4 times steeper (Bolker p. 254).
- Steeper valley = the fit gets worse faster as you move away from the MLE = the data rule out more values = a narrower interval.

Sticking point: the *height* of the minimum also changes with n (more data means a smaller

maximum likelihood, larger minimum NLL). That’s why the figure subtracts the minimum. Only *differences* in NLL between parameter values on the same data mean anything.

4. The likelihood ratio interval (~7 min)

We need a rule for “how far up the valley wall is still plausible.”

- **Informal version (Edwards, cited in Bolker p. 254):** this is the WOD’s rule of thumb. Keep every value within 2 log-likelihood units of the best. That’s any value whose likelihood is at least $1/e^2 \approx 1/7.4$ of the maximum.
- **Frequentist version (the likelihood ratio test):** the deviance, $2[\text{NLL}(\lambda) - \text{NLL}(\hat{\lambda})]$, has approximately a χ^2 distribution with 1 degree of freedom when λ is the true value. Its 95th percentile is 3.84, so the cutoff on the NLL scale is $3.84/2 = 1.92$. (This is Wilks’ theorem. Don’t derive it; Bolker’s footnote on p. 255 gives the intuition, which you can paraphrase: near the MLE the NLL is a parabola, and a squared normal deviate is χ^2_1 .)
- The 95% interval is every λ whose NLL is within 1.92 of the minimum: where the curve crosses the red line in the figure above.

```
lr_cut

## [1] 1.920729

f_cut <- function(l, y) nll_pois(l, y) - nll_pois(mean(y), y) - lr_cut
lr_int <- c(uniroot(f_cut, c(0.5, 3), y = haypiles)$root,
           uniroot(f_cut, c(3, 8), y = haypiles)$root)
lr_int_24 <- c(uniroot(f_cut, c(1, 3), y = haypiles_24)$root,
              uniroot(f_cut, c(3, 6), y = haypiles_24)$root)
round(rbind(plots_6 = lr_int, plots_24 = lr_int_24), 2)

##           [,1] [,2]
## plots_6  1.82 4.61
## plots_24 2.36 3.75
```

- With 6 plots: $\lambda \in [1.82, 4.61]$. With 24: $[2.36, 3.75]$, about half as wide (4 times the data, $\sqrt{4} = 2$ times narrower).
- The 6-plot interval is **not symmetric** around 3 (it reaches 1.61 above but only 1.18 below). That’s real: counts can’t go below zero, and the NLL curve is steeper on the low side.
- WOD callback: the chytrid curve was lopsided for the same reason, and more so, since $\hat{p} = 0.1$ sits closer to its boundary.
- Popperian reading: $\lambda = 1.5$ is falsified at the 5% level; $\lambda = 2$ survives; so does 4. The interval is the set of hypotheses that survive.

5. Return to the big picture (~1 min)

- The MLE says where the valley bottom is; the curvature says how wide it is. Width = uncertainty.
- Tomorrow's lab: the curvature shortcut (the Hessian) and what changes with more than one parameter. Wednesday: the Bayesian route.
- Next in the loop: COMPARE. The likelihood ratio test we used today to build an interval is the same test we'll use to compare models.

Discussion (~7 min)

Pick one or two:

1. Is $\lambda = 1.5$ haypiles per plot *disproven* by these data? Is $\lambda = 3$ *proven*? What would change your answer?
2. A reviewer says six plots is too few to trust this interval. Do you agree? What would you do to check?

Question 1 reinforces the survive/falsify vocabulary. (The “95% chance” confidence vs. credible question moved to Wednesday.)

PART 2: PODIUM CHEAT SHEET

Terse on purpose. Full rationale and code: Part 1.

WOD ~8 min. Lecture ~21 min. Discussion ~7 min.

WOD: Reading the Interval (~8 min)

Chytrid: 2 of 20 positive. 4 min alone, 2 with neighbor.

- 1a: $\hat{p} = 0.10$
- 1b: about 0.02 to 0.28
- 1c: lopsided (0.08 below, 0.18 above); boundary at 0
- 2a: 8 of 80, same $\hat{p}$, narrower (about 0.05 to 0.18)
- Don't mention the Wald interval here (take-home 2d)

1. Big picture (~3 min)

- Loop on board: THEORY $\rightarrow$ MODEL $\rightarrow$ DATA (fit) $\rightarrow$ COMPARE $\rightarrow$ repeat
- So far: MLE = one number. Useless without uncertainty (Bolker p. 249)
- Today: **shape of the likelihood = how sure we are**
- Same likelihood, two schools: CI (today), posterior (Wed.)
- CI = values that *survive*; outside = falsified; nothing proven

2. Reading questions (~4 min)

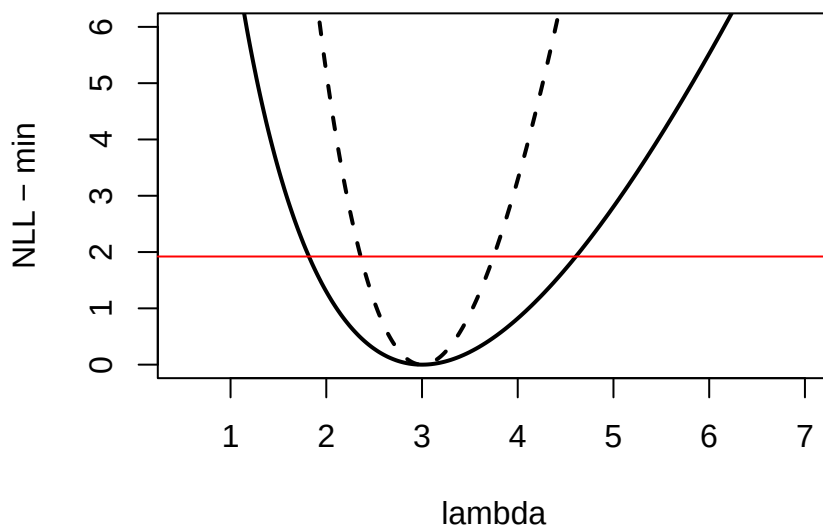
Seeds if silent:

- Why are slices “usually wrong”? (lab tomorrow)
- More data, *smaller* max likelihood, better estimate? (coin footnote)
- Where does 1.92 come from? (sec. 4)

Park answers on the board corner.

3. Curvature is information (~6 min)

Pikas: **2, 5, 3, 0, 4, 4**. Same data $\times 4 = 24$ plots.



- Callback: Lab 5 Act 1, 3a (3 vs 30 wetlands)
- NLL is a sum: $4 \times \text{data} = 4 \times \text{steeper}$
- **Sticking point:** height of minimum changes with n ; only *differences* matter

4. Likelihood ratio interval (~7 min)

- Edwards: within 2 log-lik units (the WOD rule)
- LRT: $2 \Delta \text{NLL} \sim \chi_1^2$; 95% = 3.84; cutoff **1.92**
- 6 plots: [1.82, 4.61] (asymmetric!)
- 24 plots: [2.36, 3.75] ($\sqrt{4} = 2\times$ narrower)
- $\lambda = 1.5$ falsified; 2 and 4 survive

5. Wrap (~1 min)

- Bottom of valley = estimate; width = uncertainty
- Same LRT next: COMPARE models
- Tomorrow: Hessian shortcut, slices vs. profiles. Wed.: Bayes

Discussion (~7 min)

1. Is $\lambda = 1.5$ disproven? Is 3 proven?
2. $n = 6$: trust this interval? How would you check?